ENAE 641 Midterm Formula Sheet

V. Srivastava · Fall 2026 · HW 01–03 scope

1 Linear algebra by hand

Products. $A\mathbf{x} = \sum_k x_k \mathbf{a}_k$ (a combination of the columns $\mathbf{a}_k$) $= \left[\boldsymbol{\alpha}_i^{\mathsf T}\mathbf{x}\right]_i$ (dot products with the rows $\boldsymbol{\alpha}_i^{\mathsf T}$); $(AB)_{ij} = \boldsymbol{\alpha}_i^{\mathsf T}\mathbf{b}_j$ (inner sizes match, outer give the size). $(AB)^{\mathsf T} = B^{\mathsf T}A^{\mathsf T}$, $(AB)^{-1} = B^{-1}A^{-1}$, $(A^{\mathsf T})^{-1} = (A^{-1})^{\mathsf T}$.

Det & inverse.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}, \quad A^{-1} = \frac{\operatorname{Adj}(A)}{\det(A)}$$

$\operatorname{Adj}(A)_{ij} = (-1)^{i+j}M_{ji}$ (transposed cofactors). 3×3: expand along any row/col, $\det A = \sum_j (-1)^{i+j} a_{ij} M_{ij}$ (M_{ij} = minor: delete row i, col j), or Sarrus for $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$: $aei + bfg + cdh - ceg - afh - bdi$. $\det(AB) = \det A \det B$, $\det A^{\mathsf T} = \det A$, $\det A^{-1} = 1/\det A$, $\det(cA) = c^n \det A$; triangular: product of diagonal; $\det\begin{bmatrix} A & B \\ 0 & D \end{bmatrix} = \det A \det D$; swapping rows flips the sign, adding a multiple of a row changes nothing. Blocks: $\begin{bmatrix} A & B \\ 0 & D \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1} & -A^{-1}BD^{-1} \\ 0 & D^{-1} \end{bmatrix}$, $\operatorname{blkdiag}\{A_i\}^{-1} = \operatorname{blkdiag}\{A_i^{-1}\}$.

Rank, range, nullspace (RREF). Row-reduce A ($p \times m$): rank ρ = # pivots = # independent columns = # independent rows ($\rho(A^{\mathsf T}) = \rho(A)$). $\mathcal{R}(A) \subset \mathbb{R}^p$: span of the **pivot columns of the original** A. $\mathcal{N}(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\} \subset \mathbb{R}^m$: write pivot variables in terms of free ones; one basis vector per free variable.

$\boxed{\rho(A) + \nu(A) = m \ (\#\text{ columns})}$. Square A: $\det A = 0 \Leftrightarrow \rho < n \Leftrightarrow \nu > 0 \Leftrightarrow A^{-1}$ DNE $\Leftrightarrow$ columns dependent $\Leftrightarrow \lambda = 0$ is an eigenvalue.

Span, independence, basis. Stack vectors as columns of M: independent $\Leftrightarrow \rho(M) = \#$ vectors (square: $\det M \neq 0$); $\mathbf{y} \in \operatorname{span} \Leftrightarrow \rho([M, \mathbf{y}]) = \rho(M)$. Any n independent vectors (any nonsingular matrix) form a basis of $\mathbb{R}^n$; coordinates of $\mathbf{x}$ in basis V: solve $V\mathbf{c} = \mathbf{x}$, $\mathbf{c} = V^{-1}\mathbf{x}$ (unique).

Template (HW1): $A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 2 & 0 \\ 6 & 3 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$: $\rho = 1$, $\nu = 2$, $\mathcal{R} = \operatorname{span}\left\{\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}\right\}$, $\mathcal{N} = \operatorname{span}\left\{\begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right\}$.

Eigenvalues by hand. λ with $A\mathbf{v} = \lambda\mathbf{v}$, $\mathbf{v} \neq \mathbf{0} \Leftrightarrow (\lambda I - A)$ singular:

1. Characteristic polynomial $p(\lambda) = \det(\lambda I - A) = 0$ (n roots with multiplicity).
2. 2×2: $\boxed{\lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0}$, $\lambda = \frac{\operatorname{tr}}{2} \pm \sqrt{(\operatorname{tr}/2)^2 - \det}$; complex iff $\det > (\operatorname{tr}/2)^2$: $\sigma = \operatorname{tr}/2$, $\omega = \sqrt{\det - (\operatorname{tr}/2)^2}$.
3. 3×3: $\lambda^3 - (\operatorname{tr} A)\lambda^2 + (M_{11} + M_{22} + M_{33})\lambda - \det A$ (M_{ii}: principal 2×2 minors).
4. Roots without a calculator: integer roots divide the constant term (try $\pm 1, \pm 2, \ldots$); deflate by synthetic division; then quadratic formula or complete the square $(\lambda + a)^2 + b^2 = 0 \Rightarrow \lambda = -a \pm jb$.
5. Check: $\sum \lambda_k = \operatorname{tr} A$, $\prod \lambda_k = \det A$.

Shortcuts: triangular / block-triangular / block-diagonal ⇒ diagonal entries / eigenvalues of the diagonal blocks; companion $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix}$ (or its transpose) ⇒ $\lambda^3 + a_2\lambda^2 + a_1\lambda + a_0$; $\begin{bmatrix} a & b \\ -b & a \end{bmatrix} \Rightarrow a \pm jb$; $\rho(A) = r < n \Rightarrow \lambda = 0$ ($\geq n - r$ eigvecs); constant row sum $c \Rightarrow \lambda = c$, $\mathbf{v} = [1, \ldots, 1]^{\mathsf T}$. A, $A^{\mathsf T}$, $P^{-1}AP$ share λ; $A^k, A^{-1}, A + cI, cA, e^{At} \to \lambda^k, 1/\lambda, \lambda + c, c\lambda, e^{\lambda t}$ (same $\mathbf{v}$); real A: conjugate pairs; symmetric: real λ, orthogonal $\mathbf{v}$; skew: $j\omega$ or 0.

Eigenvectors by hand.

1. For each λ solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ by RREF; one vector per free variable; # independent $= n - \rho(A - \lambda I)$ (geometric multiplicity).
2. 2×2 $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$: $\mathbf{v} = \begin{bmatrix} b \\ \lambda - a \end{bmatrix}$ or $\begin{bmatrix} \lambda - d \\ c \end{bmatrix}$. 3×3 with $\rho(A - \lambda I) = 2$: $\mathbf{v} = \mathbf{r}_i \times \mathbf{r}_j$ (cross product of two independent rows of $A - \lambda I$; fine for complex λ). Companion with 1s above the diagonal: $\mathbf{v} = [1, \lambda, \lambda^2]^{\mathsf T}$ (2×2: $[1, \lambda]^{\mathsf T}$); for its transpose that vector is the **left** eigenvector.
3. Scale is arbitrary; $\overline{\lambda} \to \overline{\mathbf{v}}$; real pair $\operatorname{Re}\mathbf{v}, \operatorname{Im}\mathbf{v}$; check $A\mathbf{v} = \lambda\mathbf{v}$.

Left eigenvectors: rows $\mathbf{w}_k^{\mathsf T}$ of V^{-1} (2×2: invert V with the formula above; $\mathbf{w}_k^{\mathsf T}A = \lambda_k \mathbf{w}_k^{\mathsf T}$, i.e. $A^{\mathsf T}\mathbf{w}_k = \lambda_k \mathbf{w}_k$), scaled so $\mathbf{w}_i^{\mathsf T}\mathbf{v}_j = \delta_{ij}$. **Repeated** λ (alg. mult. r): diagonalizable $\Leftrightarrow n - \rho(A - \lambda I) = r$ for every λ (always if distinct). Else chain $(A - \lambda I)\mathbf{p}_1 = \mathbf{0}$, $(A - \lambda I)\mathbf{p}_{k+1} = \mathbf{p}_k \Rightarrow AP = PJ$. # Jordan blocks $= n - \rho(A - \lambda I)$; # blocks of size $\geq k = \rho\big((A - \lambda I)^{k-1}\big) - \rho\big((A - \lambda I)^k\big)$.

(a): $\begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix}$: $\lambda^2 + 5\lambda + 6 = (\lambda + 2)(\lambda + 3)$; $\mathbf{v} = [1, \lambda]^{\mathsf T}$: $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$; $V^{-1} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \Rightarrow \mathbf{w}_1^{\mathsf T} = [3, 1]$, $\mathbf{w}_2^{\mathsf T} = [2, 1]$.

(b) HW2: $A = \begin{bmatrix} 0 & 0 & -13 \\ 1 & 0 & -17 \\ 0 & 1 & -5 \end{bmatrix}$: $p = \lambda^3 + 5\lambda^2 + 17\lambda + 13$, $p(-1) = 0 \Rightarrow p = (\lambda + 1)((\lambda + 2)^2 + 9)$, $\lambda = -1, -2 \pm 3j$. Rows 2, 3 of $A - \lambda I$: $\mathbf{r}_2 \times \mathbf{r}_3 = [\lambda^2 + 5\lambda + 17, \lambda + 5, 1]^{\mathsf T} \Rightarrow \mathbf{v}_{-2+3j} = [2 + 3j, 3 + 3j, 1]^{\mathsf T}$, $\mathbf{v}_{-1} = [13, 4, 1]^{\mathsf T}$.

(c) repeated: $A = \begin{bmatrix} -2 & 1 \\ -1 & -4 \end{bmatrix}$: $(\lambda + 3)^2$; $A + 3I = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$, $\rho = 1 \Rightarrow$ one block; $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $(A + 3I)\mathbf{p}_2 = \mathbf{v}_1 \Rightarrow \mathbf{p}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$; $e^{At} = e^{-3t}[I + t(A + 3I)]$.

Definiteness (symmetric M; only $M_{\text{sym}} = (M + M^{\mathsf T})/2$ enters $\mathbf{x}^{\mathsf T}M\mathbf{x}$). PD $\Leftrightarrow$ all $\lambda > 0 \Leftrightarrow$ leading principal minors > 0 (Sylvester); $\begin{bmatrix} a & b \\ b & c \end{bmatrix} \succ 0 \Leftrightarrow a > 0, ac - b^2 > 0$. PSD $\Leftrightarrow$ all $\lambda \geq 0 \Leftrightarrow$ **all** principal minors ≥ 0.

Commuting. $AB = BA$ for: diagonal pairs; A with I, A^k, A^{-1}, $p(A)$, e^{At}; $f(t)M$ with $g(\tau)M$. **Calculus.** $\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{x}^{\mathsf T}P\mathbf{x}) = \dot{\mathbf{x}}^{\mathsf T}P\mathbf{x} + \mathbf{x}^{\mathsf T}\dot{P}\mathbf{x} + \mathbf{x}^{\mathsf T}P\dot{\mathbf{x}}$; $\frac{\mathrm{d}}{\mathrm{d}t}M^{-1} = -M^{-1}\dot{M}M^{-1}$; Leibniz $\frac{\mathrm{d}}{\mathrm{d}t}\int_{a(t)}^{b(t)} f(t,\tau)\,\mathrm{d}\tau = f(t,b)\dot{b} - f(t,a)\dot{a} + \int_a^b \frac{\partial f}{\partial t}\,\mathrm{d}\tau$.

Laplace. $e^{at} \leftrightarrow \frac{1}{s-a}$, $t^k e^{at} \leftrightarrow \frac{k!}{(s-a)^{k+1}}$, $e^{\sigma t}\cos\omega t \leftrightarrow \frac{s-\sigma}{(s-\sigma)^2+\omega^2}$, $e^{\sigma t}\sin\omega t \leftrightarrow \frac{\omega}{(s-\sigma)^2+\omega^2}$, $\dot{f} \leftrightarrow sF - f(0^-)$, $\delta \leftrightarrow 1$; linear only for **constant** coefficients. Cover-up: $c_i = [(s - p_i)F(s)]_{s=p_i}$; double pole: $c_2 = [(s-p)^2F]_p$, $c_1 = \left[\frac{\mathrm{d}}{\mathrm{d}s}(s-p)^2F\right]_p$; complex pair: write $(\alpha(s-\sigma) + \beta\omega)/((s-\sigma)^2 + \omega^2)$.

Integrals. $\int e^{at}\,\mathrm{d}t = e^{at}/a$; $\int te^{at}\,\mathrm{d}t = e^{at}(at-1)/a^2$; $\int e^{at}\cos bt\,\mathrm{d}t = e^{at}(a\cos bt + b\sin bt)/(a^2+b^2)$; $\int e^{at}\sin bt\,\mathrm{d}t = e^{at}(a\sin bt - b\cos bt)/(a^2+b^2)$. **Values.** $\sin(\pi/6, \pi/4, \pi/3) = 1/2, \sqrt{2}/2, \sqrt{3}/2$ (cos reversed); $\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$, $\sqrt{5} \approx 2.236$; $\ln 2 \approx 0.693$, $\ln 50 \approx 3.91$; $e^{-1} \approx 0.368$, $e^{-4} \approx 0.018$.

2 Models, linearization, equilibria

State. $z_1, \ldots, z_n$ such that $\mathbf{z}(t_0)$ plus $\mathbf{w}(t)$, $t \geq t_0$, determine $\boldsymbol{\eta}(t)$, $t \geq t_0$. Implicit $\mathbf{0} = \mathbf{f}(\dot{\mathbf{z}}, \mathbf{z}, \mathbf{w}, t)$ (explicit $\dot{\mathbf{z}} = \mathbf{f}(\mathbf{z}, \mathbf{w}, t)$ can be too restrictive), output $\boldsymbol{\eta} = \mathbf{h}(\mathbf{z}, \mathbf{w}, t)$. Linear $\Leftrightarrow \mathbf{f} = E(t)\dot{\mathbf{z}} + F(t)\mathbf{z} + G(t)\mathbf{w}$, $\mathbf{h} = C\mathbf{z} + D\mathbf{w}$; standard form ($E$ invertible): $\boxed{A = -E^{-1}F,\ B = -E^{-1}G}$. Constant matrices ⇒ time-invariant.

Linearization about a reference solution $\mathbf{z}^*(t)$ driven by $\mathbf{w}^*(t)$: Jacobians

$$E = \left[\frac{\partial \mathbf{f}}{\partial \dot{\mathbf{z}}}\right]_*, \quad F = \left[\frac{\partial \mathbf{f}}{\partial \mathbf{z}}\right]_*, \quad G = \left[\frac{\partial \mathbf{f}}{\partial \mathbf{w}}\right]_*, \quad C = \left[\frac{\partial \mathbf{h}}{\partial \mathbf{z}}\right]_*$$

$D = \left[\frac{\partial \mathbf{h}}{\partial \mathbf{w}}\right]_*$, $E_{ij} = \frac{\partial f_i}{\partial \dot{z}_j}$, etc., evaluated at the reference **after** differentiating. Then

$$\dot{\mathbf{x}} = A(t)\mathbf{x} + B(t)\mathbf{u} + \boldsymbol{\Gamma}, \quad \mathbf{y} = C\mathbf{x} + D\mathbf{u} + \boldsymbol{\Gamma}_y$$

$\mathbf{x} = \mathbf{z} - \mathbf{z}^*$, $\mathbf{u} = \mathbf{w} - \mathbf{w}^*$, $\mathbf{y} = \boldsymbol{\eta} - \boldsymbol{\eta}^*$; $\|\boldsymbol{\Gamma}\|$ (H.O.T.) small only while $\|\mathbf{x}\|, \|\mathbf{u}\|$ small (Taylor).

! $\mathbf{x} \to \mathbf{0}$ means $\mathbf{z} \to \mathbf{z}^*$: back to the **reference**, not to 0. $\mathbf{u}$ is not the physical input: apply $\mathbf{w} = \mathbf{w}^* + \mathbf{u}$; $\mathbf{y}$ is a deviation too. Reference trajectory ⇒ LTV $A(t)$ (e.g. unicycle); equilibrium ⇒ LTI.

Equilibria / trim. $\dot{\mathbf{z}}^* = \mathbf{0}$, $\mathbf{z}^*$ const: $\mathbf{0} = \mathbf{f}(\mathbf{0}, \mathbf{z}^*, \mathbf{w}^*)$ is **algebraic**: solve $\mathbf{z}^*$ given $\mathbf{w}^*$ or vice versa. One $\mathbf{w}^*$ can give several equilibria with opposite stability (pendulum $w^* = 0$: $\theta^* = 0$ stable, π unstable).

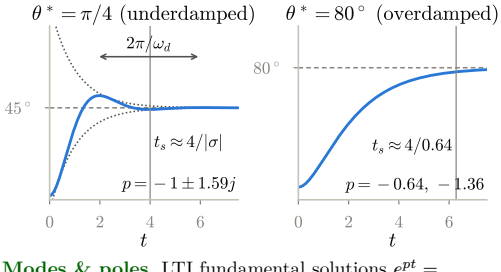
Pendulum: $ml^2\ddot{\theta} = -mgl\sin\theta - b\big(\dot{\theta}\big) + \tau_m$, $\mathbf{z} = \big[\theta, \dot{\theta}\big]^{\mathsf T}$, $w = \tau_m$:

$$E = \begin{bmatrix} 1 & 0 \\ 0 & ml^2 \end{bmatrix}, \quad F = \begin{bmatrix} 0 & -1 \\ mgl\cos\theta^* & b'\big(\dot{\theta}^*\big) \end{bmatrix}, \quad G = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

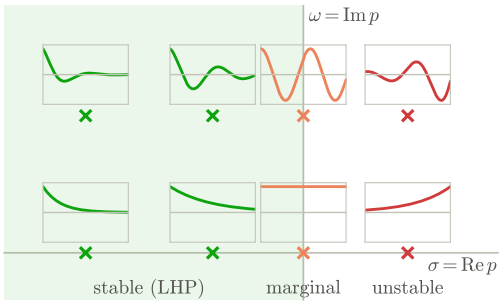
$$\Rightarrow A = \begin{bmatrix} 0 & 1 \\ -\alpha_0 & -\alpha_1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \beta_0 \end{bmatrix}$$

$\alpha_0 = (g/l)\cos\theta^*$, $\alpha_1 = b'\big(\dot{\theta}^*\big)/(ml^2)$, $\beta_0 = 1/(ml^2)$; I/O: $\ddot{y} + \alpha_1\dot{y} + \alpha_0 y = \beta_0 u$. Trim: $\boxed{mgl\sin\theta^* = w^*}$. Slide cases ($g/l = 5$, $b'(0)/(ml^2) = 2$):

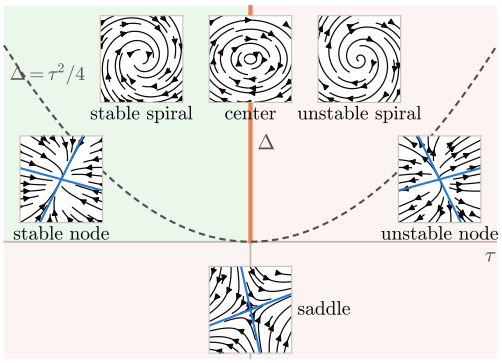
θ^*	char. poly	poles	behaviour
0	$s^2 + 2s + 5$	$-1 \pm 2j$	decaying oscillation
$\pi/4$	$s^2 + 2s + 3.54$	$-1 \pm 1.59j$	same decay, lower ω
80°	$s^2 + 2s + 0.87$	$-0.64, -1.36$	no oscillation
90°	$s^2 + 2s$	$0, -2$	**not stable**
180°	$s^2 + 2s - 5$	$1.45, -3.45$	unstable



Modes & poles. LTI fundamental solutions $e^{pt} = e^{\sigma t}(\cos\omega t + j\sin\omega t)$, $p = \sigma + j\omega$: $|e^{pt}| = e^{\sigma t} \to 0 \Leftrightarrow \operatorname{Re} p < 0$; repeated $p \Rightarrow t^\ell e^{pt}$. 2% settling: $\boxed{t_s = \ln 50/|\sigma| \approx 4/|\sigma|}$. $s^2 + 2\zeta\omega_n s + \omega_n^2$: $p = -\zeta\omega_n \pm j\omega_d$, $\omega_n = |p|$, $\boxed{\zeta = |\sigma|/|p|} = \cos(\angle \text{ from } -\mathbb{R})$, $\omega_d = \omega_n\sqrt{1 - \zeta^2}$; $\zeta < 1$ under-, $= 1$ critically, > 1 over-damped. Sketch: envelope $\pm Ae^{\sigma t}$, period $2\pi/\omega_d$, settles to the equilibrium.



2×2 type ($\tau = \operatorname{tr} A$, $\Delta = \det A$, $\lambda = \tau/2 \pm \sqrt{\tau^2/4 - \Delta}$): AS $\Leftrightarrow \tau < 0$ & $\Delta > 0$; $\Delta < 0$ saddle; $\tau = 0 < \Delta$ center; $\Delta = \tau^2/4$ repeated. $A = \begin{bmatrix} 0 & 1 \\ -k & -b \end{bmatrix}$ ($\ddot{y} + b\dot{y} + ky = 0$): AS $\Leftrightarrow k > 0, b > 0$; $k < 0$ saddle.



PD on a linearization: $\ddot{y} + \alpha_1\dot{y} + \alpha_0 y = \beta_0 u$, $u = -k_p y - k_d\dot{y} \Rightarrow s^2 + (\alpha_1 + \beta_0 k_d)s + (\alpha_0 + \beta_0 k_p)$; match $2\zeta\omega_n$, ω_n^2: $k_d = (2\zeta\omega_n - \alpha_1)/\beta_0$, $k_p = (\omega_n^2 - \alpha_0)/\beta_0$. Apply $w = w^* - k_p(\theta - \theta^*) - k_d\dot{\theta}$.

Hoop $\dot{\omega} + b\omega + a^2\sin\theta = \Omega^2\cos\theta\sin\theta$: $A = \begin{bmatrix} 0 & 1 \\ -k & -b \end{bmatrix}$, $k = a^2\cos\theta^* - \Omega^2\cos 2\theta^*$; bottom $k = a^2 - \Omega^2$, top $k = -a^2 - \Omega^2$ (saddle); $\Omega^2 > a^2$: bottom becomes a saddle, new $\cos\theta^* = a^2/\Omega^2$ with $k = \Omega^2\sin^2\theta^* > 0$.

3 Solutions: fundamental matrix & STM

Problem. $\dot{\mathbf{x}} = A(t)\mathbf{x} + B(t)\mathbf{u}$, $\mathbf{x}(t_0) = \mathbf{x}_0$; knowing n solutions from independent ICs gives all solutions.

Fundamental matrix. $\Psi = [\boldsymbol{\psi}_1, \ldots, \boldsymbol{\psi}_n]$, $\dot{\boldsymbol{\psi}}_k = A\boldsymbol{\psi}_k$, **any** independent ICs: $\dot{\Psi} = A(t)\Psi$, $\Psi(t_0)$ nonsingular. $\mathbf{x}(t) = \Psi(t)\boldsymbol{\alpha}$, $\boldsymbol{\alpha} = \Psi^{-1}(t_0)\mathbf{x}_0$ conserved when $\mathbf{u} = \mathbf{0}$. Liouville: $\frac{\mathrm{d}}{\mathrm{d}t}\det\Psi = \operatorname{tr} A(t)\det\Psi \Rightarrow \det\Psi(t) = \det\Psi(t_0)e^{\int_{t_0}^t \operatorname{tr} A\,\mathrm{d}\tau} \neq 0$ (unless $\int \operatorname{tr} A = -\infty$). **STM.** $\boxed{\Phi(t,\tau) = \Psi(t)\Psi^{-1}(\tau)}$, unique (same for every Ψ); $\mathbf{x}(t) = \Phi(t, t_0)\mathbf{x}_0$. Properties: (1) $\Phi(t,t) = I$; (2) nonsingular for all t, τ; (3) $\Phi^{-1}(t,\tau) = \Phi(\tau, t)$; (4) $\frac{\partial}{\partial t}\Phi(t,\tau) = A(t)\Phi(t,\tau)$, $\frac{\partial}{\partial \tau}\Phi(t,\tau) = -\Phi(t,\tau)A(\tau)$ (2nd argument!); (5) $\Phi(t_2, t_0) = \Phi(t_2, t_1)\Phi(t_1, t_0)$ (composition) $\Rightarrow \Phi(t,\tau) = \Phi(t,0)\Phi(\tau,0)^{-1}$. Also $\frac{\mathrm{d}}{\mathrm{d}t}\Psi^{-1} = -\Psi^{-1}A$. **General solution** (zero-input + zero-state):

$$\boxed{\mathbf{x}(t) = \underbrace{\Phi(t,t_0)\mathbf{x}_0}_{\mathbf{x}_{zi}} + \underbrace{\int_{t_0}^t \Phi(t,\tau)B(\tau)\mathbf{u}(\tau)\,\mathrm{d}\tau}_{\mathbf{x}_{zs}}}$$

$\mathbf{y} = C(t)\Phi(t,t_0)\mathbf{x}_0 + \int_{t_0}^t G(t,\tau)\mathbf{u}(\tau)\,\mathrm{d}\tau$, impulse response $\boxed{G(t,\tau) = C(t)\Phi(t,\tau)B(\tau) + D(t)\delta(t-\tau)}$ ($\equiv$ using $D(\tau)\delta(t-\tau)$; $p \times m$; $\int f(\tau)\delta(t-\tau)\,\mathrm{d}\tau = f(t)$). View: $\mathbf{x} = \Psi\boldsymbol{\alpha}(t)$, $\dot{\boldsymbol{\alpha}} = \Psi^{-1}B\mathbf{u}$. **Peano–Baker.** $\Phi = I + \int_\tau^t A(\sigma_1)\,\mathrm{d}\sigma_1 + \int_\tau^t A(\sigma_1)\int_\tau^{\sigma_1} A(\sigma_2)\,\mathrm{d}\sigma_2\,\mathrm{d}\sigma_1 + \ldots$

$\boxed{\text{If } A(t)M = MA(t),\ M(t,\tau) = \int_\tau^t A(\sigma)\,\mathrm{d}\sigma, \text{ then } \Phi(t,\tau) = e^{M(t,\tau)} = \sum_k M^k/k!}$

! Only when $A(t)$ commutes with its integral; in general $\Phi \neq \exp\int A$.

Commuting sum $A(t) = \sum_j f_{j(t)}A_j$, $A_iA_j = A_jA_i$: $\Phi = \prod_j e^{A_j\int_\tau^t f_j(\sigma)\,\mathrm{d}\sigma}$. **Block-triangular** $\begin{bmatrix} A_1 & A_{12} \\ 0 & A_2 \end{bmatrix} \Rightarrow \Phi = \begin{bmatrix} \Phi_1 & \Phi_{12} \\ 0 & \Phi_2 \end{bmatrix}$, $\Phi_{12}(t,\tau) = \int_\tau^t \Phi_1(t,\sigma)A_{12}(\sigma)\Phi_2(\sigma,\tau)\,\mathrm{d}\sigma$. **Scalar:** $\dot{x} = a(t)x + b(t)u \Rightarrow x = e^{\int_{t_0}^t a}x_0 + \int_{t_0}^t e^{\int_\sigma^t a}b(\sigma)u(\sigma)\,\mathrm{d}\sigma$.

Coordinates $\mathbf{x} = P(t)\mathbf{z}$: $\Phi_z = P(t)\Phi(t,\tau)P^{-1}(\tau)$. **LTI.** $\Phi = e^{A(t-\tau)}$ (elapsed time only $\Rightarrow t_0 = 0$):

$$\boxed{\mathbf{x} = e^{At}\mathbf{x}_0 + \int_0^t e^{A(t-\tau)}B\mathbf{u}(\tau)\,\mathrm{d}\tau}, \quad G(t) = Ce^{At}B + D\delta(t).$$

Building Φ for LTV by hand: (i) triangular/decoupled rows: solve the scalar row, substitute upward; (ii) check commutativity ⇒ exponential of the integral; (iii) split $A = \sum f_jA_j$; (iv) find $P(t)$ making A_p constant (e.g. a

rotation); (v) periodic $A(t)$: Floquet; or any n independent solutions $\Rightarrow \Phi = \Psi(t)\Psi^{-1}(\tau)$.

Triangular (HW2): $\dot{x}_1 = tx_2$, $\dot{x}_2 = -x_2$: $x_2 = e^{-(t-\tau)}x_2(\tau)$, $x_1 = x_1(\tau) + \int_\tau^t \sigma e^{-(\sigma-\tau)}\,\mathrm{d}\sigma\, x_2(\tau) \Rightarrow \Phi = \begin{bmatrix} 1 & (\tau+1)-(t+1)e^{-(t-\tau)} \\ 0 & e^{-(t-\tau)} \end{bmatrix}$.

Commuting (HW2): $\begin{bmatrix} \cos 3t & \sin 3t \\ -\sin 3t & \cos 3t \end{bmatrix} = \cos 3t\, I + \sin 3t\, S$, $S = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, $S^2 = -I$: $\Phi = e^{(\sin 3t - \sin 3\tau)/3}\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$, $\theta = (\cos 3\tau - \cos 3t)/3$.

4 Matrix exponential, resolvent, TFM

Definition. $e^{At} = \sum_{k=0}^{\infty} \frac{t^k}{k!}A^k = \Phi(t,0)$. $e^{A0} = I$; nonsingular; $e^{A(t_1+t_2)} = e^{At_1}e^{At_2}$; $\left(e^{At}\right)^{-1} = e^{-At}$; $\left(e^{At}\right)^{\mathsf{T}} = e^{A^{\mathsf{T}}t}$; $\frac{\mathrm{d}}{\mathrm{d}t}e^{At} = Ae^{At} = e^{At}A$; $e^{P^{-1}APt} = P^{-1}e^{At}P$; eigenvalues $e^{\lambda_k t}$.

! $e^M \neq [e^{m_{ij}}]$ (not entrywise). $e^{(A+B)t} = e^{At}e^{Bt}$ **only if** $AB = BA$.

A	e^{At}
$\mathrm{diag}\{a_k\}$	$\mathrm{diag}\{e^{a_k t}\}$
$\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$	$e^{at}\begin{bmatrix} \cos bt & \sin bt \\ -\sin bt & \cos bt \end{bmatrix}$
$\begin{bmatrix} a & 1 \\ 0 & a \end{bmatrix}$	$e^{at}\begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$
$J_3(a)$	$e^{at}\begin{bmatrix} 1 & t & t^2/2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{bmatrix}$; J_n: $t^k/k!$ on superdiag. k
$\mathrm{blkdiag}\{A_i\}$	$\mathrm{blkdiag}\{e^{A_i t}\}$
θS, $S^2 = -I$	$\cos\theta\, I + \sin\theta\, S$
N, $N^k = 0$	$I + Nt + \ldots + N^{k-1}t^{k-1}/(k-1)!$ (finite)

2×2 closed forms. Distinct real $\lambda_{1,2}$: $e^{At} = \frac{\lambda_1 e^{\lambda_2 t} - \lambda_2 e^{\lambda_1 t}}{\lambda_1 - \lambda_2}I + \frac{e^{\lambda_1 t} - e^{\lambda_2 t}}{\lambda_1 - \lambda_2}A$; repeated λ: $e^{\lambda t}[I + t(A - \lambda I)]$; complex $\sigma + j\omega$: $\boxed{e^{\sigma t}\left[\cos\omega t\, I + \frac{\sin\omega t}{\omega}(A - \sigma I)\right]}$. **Methods.**

1. **Resolvent:** $\boxed{e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}}$, $(sI - A)^{-1} = \mathrm{Adj}(sI - A)/\det(sI - A) = Q(s)/r(s)$: every entry has the **same** denominator $r(s)$ (degree n; numerators degree $\leq n-1$) $\Rightarrow$ partial fractions $\Rightarrow$ every entry is a combination of the same $e^{\lambda_k t}$: poles = eigenvalues; repeated $\Rightarrow t^\ell e^{\lambda t}$.
2. **Cayley–Hamilton:** $p_A(A) = 0 \Rightarrow A^{k\geq n}$ is a combination of $I, \ldots, A^{n-1} \Rightarrow e^{At} = \sum_{k=0}^{n-1}\alpha_k(t)A^k$, $\sum_k \alpha_k \lambda_i^k = e^{\lambda_i t}$ (Vandermonde, $\boldsymbol{\lambda}$ **distinct**; repeated: add $\frac{\mathrm{d}}{\mathrm{d}\lambda}$: $\sum_k k\alpha_k\lambda^{k-1} = te^{\lambda t}$). Mostly a theory tool.
3. **Diagonalize:** $A^k = V\Lambda^k V^{-1} \Rightarrow$ $\boxed{e^{At} = Ve^{\Lambda t}V^{-1} = \sum_k e^{\lambda_k t}\mathbf{v}_k\mathbf{w}_k^{\mathsf{T}}}$ (dyads $\mathbf{v}_k\mathbf{w}_k^{\mathsf{T}}$, $n \times n$).
4. **Jordan / block modal:** $e^{At} = Pe^{Jt}P^{-1}$, block by block.

Two ways: $A = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix}$: $(sI - A)^{-1} = \frac{1}{(s+2)(s+3)}\begin{bmatrix} s+5 & 1 \\ -6 & s \end{bmatrix}$, e.g. $\frac{s+5}{(s+2)(s+3)} = \frac{3}{s+2} - \frac{2}{s+3}$. Dyads (eigs from §1a): $\mathbf{v}_1\mathbf{w}_1^{\mathsf{T}} = \begin{bmatrix} 3 & 1 \\ -6 & -2 \end{bmatrix}$, $\mathbf{v}_2\mathbf{w}_2^{\mathsf{T}} = \begin{bmatrix} -2 & -1 \\ 6 & 3 \end{bmatrix} \Rightarrow e^{At} = e^{-2t}\begin{bmatrix} 3 & 1 \\ -6 & -2 \end{bmatrix} + e^{-3t}\begin{bmatrix} -2 & -1 \\ 6 & 3 \end{bmatrix}$.

TFM. $(sI - A)\mathbf{X} = \mathbf{x}_0 + B\mathbf{U}$, $\mathbf{Y} = C(sI - A)^{-1}\mathbf{x}_0 + G(s)\mathbf{U}$:

$$\boxed{G(s) = C(sI - A)^{-1}B + D} \quad (p \times m)$$

$G(t) = \mathcal{L}^{-1}\{G(s)\} = Ce^{At}B + D\delta(t)$; $[G(s)]_{ij} = Y_i/U_j$ (zero IC, other inputs 0); modal partial fractions $\boxed{[G(s)]_{ij} = \sum_{k=1}^{n}\frac{(\mathbf{c}_i^{\mathsf{T}}\mathbf{v}_k)(\mathbf{w}_k^{\mathsf{T}}\mathbf{b}_j)}{s - \lambda_k} + d_{ij}}$ invariant under $\mathbf{x} = P\mathbf{z}$.

! Laplace is LTI only: $\mathcal{L}\{A(t)\mathbf{x}(t)\} \neq A(t)\mathbf{X}(s)$ (nor $\mathcal{L}\{A\}\mathbf{X}$).

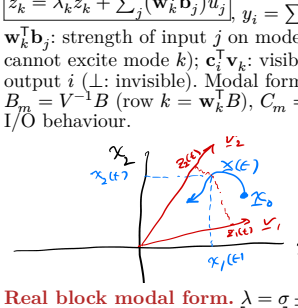
5 Modal forms, coordinates, Jordan, Floquet

Eigenstructure. $AV = V\Lambda$; distinct $\lambda \Rightarrow$ independent $\mathbf{v}_k \Rightarrow V^{-1}$ exists, $A = V\Lambda V^{-1}$. $V^{-1} = \begin{bmatrix} \mathbf{w}_1^{\mathsf{T}} \\ \vdots \\ \mathbf{w}_n^{\mathsf{T}} \end{bmatrix}$: left eigenvectors ($A^{\mathsf{T}}$ has the same λ); $\boxed{\mathbf{w}_i^{\mathsf{T}}\mathbf{v}_j = \delta_{ij}}$ (bi-orthonormal). **Modal response.**

$\boxed{\mathbf{x}(t) = \sum_k (\mathbf{w}_k^{\mathsf{T}}\mathbf{x}_0)e^{\lambda_k t}\mathbf{v}_k}$: modes $e^{\lambda_k t}\mathbf{v}_k$, weights $\mathbf{w}_k^{\mathsf{T}}\mathbf{x}_0$ = how much the IC excites mode k. $\mathbf{x}_0 = \beta\mathbf{v}_j \Rightarrow \mathbf{x} = \beta e^{\lambda_j t}\mathbf{v}_j$: only mode j, straight along the $\mathbf{v}_j$ line. **Forced, decoupled:** $\mathbf{x} = \sum_k z_k\mathbf{v}_k = V\mathbf{z}$ with

$\boxed{\dot{z}_k = \lambda_k z_k + \sum_j (\mathbf{w}_k^{\mathsf{T}}\mathbf{b}_j)u_j}$, $y_i = \sum_k (\mathbf{c}_i^{\mathsf{T}}\mathbf{v}_k)z_k + \mathbf{d}_i^{\mathsf{T}}\mathbf{u}$.

$\mathbf{w}_k^{\mathsf{T}}\mathbf{b}_j$: strength of input j on mode k ($\mathbf{w}_k \perp \mathbf{b}_j$: input j cannot excite mode k); $\mathbf{c}_i^{\mathsf{T}}\mathbf{v}_k$: visibility of mode k in output i ($\perp$: invisible). Modal form: $A_m = V^{-1}AV = \Lambda$, $B_m = V^{-1}B$ (row k = $\mathbf{w}_k^{\mathsf{T}}B$), $C_m = CV$, $D_m = D$; same I/O behaviour.



Real block modal form. $\lambda = \sigma \pm j\omega$ with $\mathbf{v}$ for $\sigma + j\omega$: $A[\mathrm{Re}\,\mathbf{v}, \mathrm{Im}\,\mathbf{v}] = [\mathrm{Re}\,\mathbf{v}, \mathrm{Im}\,\mathbf{v}]\begin{bmatrix} \sigma & \omega \\ -\omega & \sigma \end{bmatrix}$; $P = [\mathrm{Re}\,\mathbf{v}_1, \mathrm{Im}\,\mathbf{v}_1, \mathrm{Re}\,\mathbf{v}_3, \mathrm{Im}\,\mathbf{v}_3, \mathbf{v}_5, \ldots] \Rightarrow A_p = \mathrm{blkdiag}\left(\begin{bmatrix} \sigma_1 & \omega_1 \\ -\omega_1 & \sigma_1 \end{bmatrix}, \ldots, \lambda_5, \ldots\right)$, $e^{A_p t}$ = rotation blocks. **One oscillatory mode:** $\mathbf{x}_0 = a\,\mathrm{Re}\,\mathbf{v} + b\,\mathrm{Im}\,\mathbf{v} \Rightarrow z_1 = e^{\sigma t}(a\cos\omega t + b\sin\omega t)$, $z_2 = e^{\sigma t}(b\cos\omega t - a\sin\omega t)$, $\mathbf{x} = z_1\,\mathrm{Re}\,\mathbf{v} + z_2\,\mathrm{Im}\,\mathbf{v}$: only that frequency, and $\mathbf{x}(t) \in \mathrm{span}\{\mathrm{Re}\,\mathbf{v}, \mathrm{Im}\,\mathbf{v}\}$ for all t (a plane). Real mode: $\mathbf{x}_0 \propto \mathbf{v}$. General IC: $\mathbf{z}_0 = P^{-1}\mathbf{x}_0$. Polar $v_i = r_i e^{j\varphi_i}$: x_i oscillates with relative amplitude r_i and phase φ_i. Which input excites a mode: that mode's rows of $B_p = P^{-1}B$.

HW2 (eigs from §1b): $P = [\mathrm{Re}\,\mathbf{v}, \mathrm{Im}\,\mathbf{v}, \mathbf{v}_3] = \begin{bmatrix} 2 & 3 & 13 \\ 3 & 3 & 4 \\ 1 & 0 & 1 \end{bmatrix}$, $A_p = P^{-1}AP = \begin{bmatrix} -2 & 3 & 0 \\ -3 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$; $\mathbf{x}_0 = [2, 3, 1]^{\mathsf{T}}$ excites only the oscillation, $[13, 4, 1]^{\mathsf{T}}$ only e^{-t}.

Coordinate change $\mathbf{x} = P\mathbf{z}$: $A_p = P^{-1}AP$, $B_p = P^{-1}B$, $C_p = CP$, $D_p = D$; time-varying $\boxed{A_p(t) = P^{-1}\left(AP - \dot{P}\right)}$ (most general). Invariant: $\Psi_x = P\Psi_z$, $\Phi_x(t,\tau) = P(t)\Phi_z(t,\tau)P^{-1}(\tau)$, $G(t,\tau)$, $G(s)$ (insert $I = PP^{-1}$), eigenvalues (constant P). **Jordan form.** $\det(sI - A) = (s - \lambda_1)^r \ldots$: $P^{-1}AP = \mathrm{blkdiag}\{J_1(\lambda_1), \ldots, J_\ell(\lambda_1), \lambda_{r+1}, \ldots\}$, $J_i = \begin{bmatrix} \lambda & 1 & & \\ & \ddots & 1 & \\ & & \lambda & \end{bmatrix}_{n_i \times n_i}$, $\sum n_i = r$ (algebraic),

$\boxed{\ell = \nu(\lambda_1 I - A) = n - \rho(\lambda_1 I - A)}$ (geometric = # eigvecs = # blocks). $\ell = r$: diagonal; $\ell = 1$: one $r \times r$ block; $1 < \ell < r$: sizes not fixed ($r = 4$: $\ell = 3 \Rightarrow (2,1,1)$; $\ell = 2 \Rightarrow (3,1)$ or $(2,2)$), set by chains $(\lambda I - A)^k\mathbf{w} = 0$. Largest block $n_i \Rightarrow$ terms up to $t^{n_i - 1}e^{\lambda t}$. Very sensitive: $\begin{bmatrix} 1 & 1 \\ \varepsilon & 1 \end{bmatrix}$ is one block at $\varepsilon = 0$, diagonal for $\varepsilon \neq 0$.

HW2 Jordan: $(\lambda^2 + 2\lambda + 10)(\lambda + 3)^3$, $\rho(A + 3I) = 4 \Rightarrow \ell = 1$, one 3×3 block: $P^{-1}e^{At}P = \mathrm{blkdiag}\left(e^{-t}\begin{bmatrix} \cos 3t & \sin 3t \\ -\sin 3t & \cos 3t \end{bmatrix}, e^{-3t}\begin{bmatrix} 1 & t & t^2/2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{bmatrix}\right)$; $\mathbf{x}_0 = \mathbf{p}_4 \Rightarrow P^{-1}\mathbf{x} = [0, 0, te^{-3t}, e^{-3t}, 0]^{\mathsf{T}}$.

Lyapunov transformation. Any $A(t) \to$ any constant R via $P(t) = \Psi(t)e^{-Rt}$ ($A_p = R$), so $\Phi = P(t)e^{R(t-\tau)}P^{-1}(\tau)$. But $\dot{x} = x \to \dot{z} = -z$ with $P = e^{2t}$ ("unstable made stable": $P^{-1} \to 0$). Lyapunov transformation: also $|\det P(t)| \geq \eta > 0$ and $|\det P^{-1}(t)| \geq \eta\ \forall t \Rightarrow$ preserves convergence / divergence. **Floquet** ($A(t+T) = A(t)$): monodromy $\boxed{Q = \Phi(t_0 + T, t_0)}$, $R = \frac{1}{T}\log Q$ ($e^{RT} = Q$), $P(t) = \Phi(t, t_0)e^{-Rt}$ is T-periodic (a Lyapunov transformation) since $\Phi(t+T, \tau+T) = \Phi(t,\tau)$: $\boxed{\Phi(t,\tau) = P(t)e^{R(t-\tau)}P^{-1}(\tau)}$ (periodic part × LTI part)

AS $\Leftrightarrow \mathrm{Re}\,\lambda(R) < 0 \Leftrightarrow |\mu| < 1$ for all multipliers $\mu = \lambda(Q)$; any $|\mu| > 1$: unstable. $\log Q = V\,\mathrm{diag}(\ln\mu_k)V^{-1}$; $\mu < 0 \Rightarrow \ln\mu = \ln|\mu| + j\pi$ (complex R; or use period $2T$).

! Frozen-time eigenvalues of $A(t)$ do **not** decide LTV stability: HW2's periodic $A(t)$ has $\lambda = -1/4 \pm j\sqrt{7}/4$ for all t, yet $\mathbf{x} = e^{t/2}[\cos t, -\sin t]^{\mathsf{T}}$.

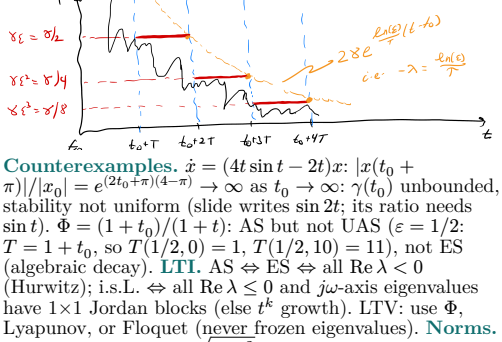
6 Stability & Lyapunov

Definitions ($\mathbf{x} = \Phi(t, t_0)\mathbf{x}_0$).

- **Stable i.s.L.:** $\exists\gamma(t_0) < \infty$: $\|\mathbf{x}(t)\| \leq \gamma\|\mathbf{x}_0\|\ \forall t \geq t_0 \Leftrightarrow \|\Phi(t, t_0)\|_i$ bounded; **uniform** if γ is independent of t_0.
- **AS:** i.s.L. **and** $\|\mathbf{x}\| \to 0$ ($\Leftrightarrow \|\Phi\| \to 0$): $\forall\varepsilon\ \exists T(\varepsilon, t_0)$: $\|\mathbf{x}(t)\| \leq \varepsilon\|\mathbf{x}_0\|\ \forall t \geq t_0 + T$. i.s.L. is needed: a finite escape time could blow up and then decay. **UAS:** T independent of t_0 (still depends on ε).
- **ES:** $\boxed{\|\mathbf{x}(t)\| \leq \gamma e^{-\lambda(t-t_0)}\|\mathbf{x}_0\|}$, $\lambda > 0$ ($\|\mathbf{x}\|$ may wiggle under the envelope); **UES:** γ, λ independent of t_0.

UES $\Rightarrow$ ES $\Rightarrow$ AS $\Rightarrow$ i.s.L.; AS $\not\Rightarrow$ ES.

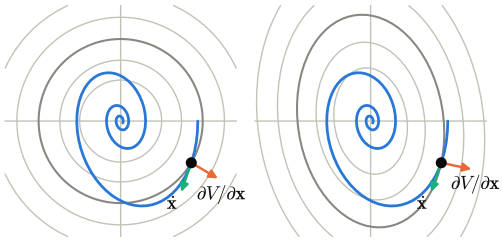
Linear: UAS $\Leftrightarrow$ UES: $\|\Phi(t_0 + T, t_0)\| \leq \varepsilon\ \forall t_0 \Rightarrow \|\mathbf{x}\| \leq \gamma\varepsilon^k\|\mathbf{x}_0\|$ on $[t_0 + kT, t_0 + (k+1)T] \Rightarrow \lambda = -\ln\varepsilon/T$.



Counterexamples. $\dot{x} = (4t\sin t - 2t)x$: $|x(t_0 + \pi)|/|x_0| = e^{(2t_0+\pi)(4-\pi)} \to \infty$ as $t_0 \to \infty$: $\gamma(t_0)$ unbounded, stability not uniform (slide writes $\sin 2t$; its ratio needs $\sin t$). $\dot{\Phi} = (1 + t_0)/(1 + t)$: AS but not UAS ($\varepsilon = 1/2$: $T = 1 + t_0$, so $T(1/2, 0) = 1$, $T(1/2, 10) = 11$), not ES (algebraic decay). **LTI.** AS $\Leftrightarrow$ ES $\Leftrightarrow$ all $\mathrm{Re}\,\lambda < 0$ (Hurwitz); i.s.L. $\Leftrightarrow$ all $\mathrm{Re}\,\lambda \leq 0$ and $j\omega$-axis eigenvalues have 1×1 Jordan blocks (else t^k growth). LTV: use Φ, Lyapunov, or Floquet (never frozen eigenvalues). **Norms.** $\|\mathbf{x}\|_1 = \sum|x_k|$, $\|\mathbf{x}\|_2 = \sqrt{\sum x_k^2}$, $\|\mathbf{x}\|_\infty = \max|x_k|$, $\|\mathbf{x}\|_p = \left(\sum|x_k|^p\right)^{1/p}$; $\|\mathbf{x}\| \geq 0$, $= 0 \Leftrightarrow \mathbf{x} = 0$, $\|\alpha\mathbf{x}\| = |\alpha|\|\mathbf{x}\|$, triangle. Equivalence: $c\|\mathbf{x}\|_q \leq \|\mathbf{x}\|_p \leq \bar{c}\|\mathbf{x}\|_q$ (stability is norm-independent). Induced $\|A\|_i = \max\|A\mathbf{x}\|/\|\mathbf{x}\|$, $\boxed{\|A\mathbf{x}\| \leq \|A\|_i\|\mathbf{x}\|}$: $\|A\|_{i,2} = \sqrt{\lambda_{\max}(A^{\mathsf{T}}A)}$, $\|A\|_{i,1} =$ max abs. column sum, $\|A\|_{i,\infty} =$ max abs. row sum; $\|A + B\| \leq \|A\| + \|B\|$, $\|AB\| \leq \|A\|\|B\|$. **Quadratic forms.** $\mathbf{x}^{\mathsf{T}}M\mathbf{x} = \mathbf{x}^{\mathsf{T}}M_{\mathrm{sym}}\mathbf{x}$ since $\mathbf{x}^{\mathsf{T}}M_{\mathrm{skew}}\mathbf{x} = 0$; Rayleigh–Ritz (symmetric M): $\boxed{\lambda_{\min}(M)\|\mathbf{x}\|_2^2 \leq \mathbf{x}^{\mathsf{T}}M\mathbf{x} \leq \lambda_{\max}(M)\|\mathbf{x}\|_2^2}$. $M(t)$ PD (PSD): $\mathbf{x}^{\mathsf{T}}M\mathbf{x} > 0$ (≥ 0) $\forall t$; uniformly PD: $\min_t \lambda_{\min}(M(t)) \geq \eta > 0$. **First test** ($P = I$): $\frac{\mathrm{d}}{\mathrm{d}t}\|\mathbf{x}\|_2^2 = \mathbf{x}^{\mathsf{T}}(A^{\mathsf{T}} + A)\mathbf{x} \leq \alpha(t)\|\mathbf{x}\|_2^2$, $\alpha = \lambda_{\max}(A^{\mathsf{T}} + A)$; $\boxed{\max_t \lambda_{\max}(A^{\mathsf{T}}(t) + A(t)) \leq 0 \Rightarrow \text{i.s.L.}}$ ($\gamma = 1$).

! Sufficient only: $\lambda_{\max} > 0$ does **not** imply instability (too restrictive: $\|\mathbf{x}\|$ may rise and still decay). Skew $A(t)$ ($A^{\mathsf{T}} = -A$) $\Rightarrow \|\mathbf{x}\|_2$ constant: uniformly i.s.L., not AS (HW3).

Lyapunov functions. Nested closed level sets $V(\mathbf{x}) = c$; $\dot{V} = \frac{\partial V}{\partial \mathbf{x}} \cdot \dot{\mathbf{x}} \leq 0$ (outward normal · velocity) $\Rightarrow$ motion points inward $\Rightarrow S(c) = \{\mathbf{x} : V \leq c\}$ is **invariant** $\Rightarrow$ i.s.L. Any nested closed contours work; linear systems: ellipses, QLF $V = \mathbf{x}^{\mathsf{T}}P\mathbf{x}$, $P \succ 0$ ($P = I$ is $\|\mathbf{x}\|^2$).



Lyapunov equation. $P(t)$ bounded, uniformly PD: $\rho = \max_t \lambda_{\max}(P)$, $\eta = \min_t \lambda_{\min}(P)$:

$$\boxed{-Q(t) = \dot{P} + A^{\mathsf{T}}P + PA} \quad \Rightarrow \quad \dot{V} = -\mathbf{x}^{\mathsf{T}}Q\mathbf{x}$$

Theorem: $Q \succeq 0 \Rightarrow$ i.s.L. ($\eta\|\mathbf{x}\|^2 \leq V(t) \leq V(t_0) \leq \rho\|\mathbf{x}_0\|^2$, $\gamma = \sqrt{\rho/\eta}$). Q uniformly PD, $\nu = \min_t \lambda_{\min}(Q) > 0 \Rightarrow$ ES: $\dot{V} \leq -\nu\|\mathbf{x}\|^2 \leq -(\nu/\rho)V$ and, by comparison, $\boxed{\|\mathbf{x}(t)\|_2 \leq \sqrt{\frac{\rho}{\eta}}e^{-\frac{\nu}{2\rho}(t-t_0)}\|\mathbf{x}_0\|_2}$

Failed test $\Rightarrow$ **inconclusive** (try another P). Comparison lemma: $\dot{v} \leq f(v,t)$, $v(t_0) \leq w(t_0)$, $\dot{w} = f(w,t) \Rightarrow v(t) \leq w(t)$. **Converse.** $A(t)$ UES $\Rightarrow$ for **any** bounded unif. PD Q there is a bounded unif. PD $P(t) = \int_t^\infty \Phi^{\mathsf{T}}(\tau,t)Q(\tau)\Phi(\tau,t)\,\mathrm{d}\tau$ ($\|P\| \leq \gamma_Q c^2/(2\nu)$); one pair $\Rightarrow$ infinitely many. **LTI:** $\boxed{A^{\mathsf{T}}P + PA = -Q}$: A Hurwitz $\Rightarrow \forall Q \succ 0\ \exists P \succ 0$.

! Does **not** say every $P \succ 0$ gives $Q \succ 0$: $P = I$ (the $\lambda(A + A^{\mathsf{T}})$ test) can fail for a stable non-normal A (HW3 F16); infinitely many other P work.

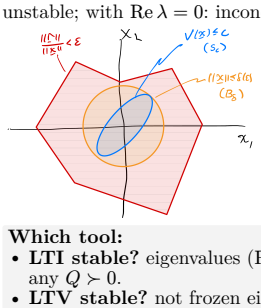
2×2 ALE: $A = \begin{bmatrix} 0 & 1 \\ -k & -b \end{bmatrix}$, $Q = I$, $P = \begin{bmatrix} p_1 & p_2 \\ p_2 & p_3 \end{bmatrix}$: (1,1) $-2kp_2 = -1$; (1,2) $p_1 - bp_2 - kp_3 = 0$; (2,2) $2(p_2 - bp_3) = -1 \Rightarrow p_2 = \frac{1}{2k}$, $p_3 = \frac{1+k}{2bk}$, $p_1 = \frac{1+k}{2b} + \frac{b}{2k}$; $P \succ 0 \Leftrightarrow k, b > 0$; then $\rho = \lambda_{\max}(P)$, $\eta = \lambda_{\min}(P)$, $\nu = 1$.

Sign-changing $\nu(t)$ (HW3): $\dot{V} \leq -\nu(t)\|\mathbf{x}\|^2 \Rightarrow \dot{V} \leq -\beta V$, $\beta = \nu^+/\rho - \nu^-/\eta$, $\nu^\pm = \max\{\pm\nu, 0\}$ (lecture: $V \leq V_0 e^{-\frac{1}{\rho}\int\alpha}$, valid where $\alpha \geq 0$) $\Rightarrow \|\mathbf{x}\| \leq \sqrt{\rho/\eta}e^{-\frac{1}{2}\int_{t_0}^t\beta}\|\mathbf{x}_0\|$. $\int\beta$ bounded below (e.g. $\int\nu^- < \infty$) $\Rightarrow$ i.s.L.; $\int\beta \to \infty \Rightarrow$ AS; $\int_t^{t+T}\beta \geq \mu > 0\ \forall t \Rightarrow$ UES. $\nu = 1/t$: AS only; $-ce^{-t}$: i.s.L.; $1/t^2$: i.s.L. only ($\int\nu < \infty$); $2 - 3\sin t$: UES if $\rho/\eta \lesssim 12.3$. $\dot{V} = \alpha V$, $\alpha > 0$, gives $V \to \infty$ only if $\int\alpha = \infty$.

Local stability. $\dot{\mathbf{x}} = A(t)\mathbf{x} + \boldsymbol{\Gamma}(\mathbf{x}, t)$, $\boldsymbol{\Gamma}$ 2nd order: $\forall\varepsilon\ \exists\delta(\varepsilon)$: $\|\boldsymbol{\Gamma}\| \leq \varepsilon\|\mathbf{x}\|$ on $B_\delta = \{\|\mathbf{x}\| \leq \delta\}$. With (P, Q) of the linear part, $\gamma_p = \max_t \|P\|_i$:

$$\dot{V} \leq -\nu\|\mathbf{x}\|^2 + 2\gamma_p\|\mathbf{x}\|\|\boldsymbol{\Gamma}\| \leq -(\nu - 2\gamma_p\varepsilon)\|\mathbf{x}\|^2 = -\nu_1\|\mathbf{x}\|^2$$

in B_δ if $\varepsilon < \nu/(2\gamma_p)$. Pick c with $S_c = \{V \leq c\} \subset B_\delta$ (e.g. $c = \eta\delta^2$; then $S_c \supset B_{\delta\sqrt{\eta/\rho}}$): S_c invariant, $\mathbf{x} \to 0$ exponentially from any $\mathbf{x}_0 \in S_c$. **Local**, not global; S_c is a conservative estimate of the region of attraction; needs a UES linearization. Linearization with some $\mathrm{Re}\,\lambda > 0$: unstable; with $\mathrm{Re}\,\lambda = 0$: inconclusive. Physically $\mathbf{z} \to \mathbf{z}^*$.



Which tool:

- **LTI stable?** eigenvalues ($\mathrm{Re}\,\lambda < 0$), or the ALE with any $Q \succ 0$.
- **LTV stable?** not frozen eigenvalues: bound $\|\Phi\|$, a Lyapunov $P(t)$, or Floquet multipliers (periodic).
- **e^{At}:** 2×2 closed forms; dyads $Ve^{\Lambda t}V^{-1}$; resolvent + partial fractions.
- **One mode:** $\mathbf{x}_0$ in its eigen-subspace; input j reaches mode k iff $\mathbf{w}_k^{\mathsf{T}}\mathbf{b}_j \neq 0$.
- **Nonlinear equilibrium:** trim $\to$ Jacobians $\to \lambda(A)$: local only, $\mathbf{z} \to \mathbf{z}^*$.